PARTH JAIN

IIT Madras B.S. Degree

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SUMMARY

Computer Science and Data Science student with experience building scalable full-stack applications, backend systems, and AI/ML solutions. Experienced in Flask, Redis, Celery, SQLAlchemy, NLP, and Hugging Face Transformers, with hands-on experience developing production-ready REST APIs, semantic search pipelines, and LLM-powered applications.

EDUCATION

Indian Institute of Technology, Madras — Chennai, Tamil Nadu
BS in Data Science and Applications — *May 2023 – May 2027*

Vellore Institute of Technology (VIT Bhopal) — Bhopal, Madhya Pradesh
BTech in Computer Science and Engineering — *Jun 2023 – Jun 2027*

EXPERIENCE

Hugging Face | *Open Source Contributor* — *Remote*

- Deployed and open-sourced AI pipelines on Hugging Face Spaces, including a Resume Analyzer & Matcher built with Sentence Transformers (BGE), and a custom NER tool using BERT.
- Engineered a highly optimized two-tier semantic matching system leveraging FlashText for O(N) taxonomy skill mapping and KeyBERT for dynamic gap-filling, outperforming standard regex-based parsers.
- Contributed to the AI open-source community by exposing model endpoints as REST APIs, demonstrating practical implementations of modern NLP architectures for production environments.

PROJECTS

Placement Portal System | *Flask, Vue.js, SQLite, SQLAlchemy, Redis, Celery, JWT*

- Engineered a scalable multi-role placement platform featuring 40+ REST APIs covering student, recruiter, and admin workflows, reducing administrative manual effort by an estimated 30%.
- Architected a robust JWT-based Role-Based Access Control (RBAC) system, ensuring 100% data isolation between student and recruiter accounts with secure session management and token handling.
- Achieved a 40% reduction in API response latency and a 25% boost in responsiveness under concurrent loads by integrating Redis caching and Celery for asynchronous background task processing.

Machine Learning Text Classification Pipeline | *Python, Scikit-learn, NLP, TF-IDF, Ensembles*

- Constructed an end-to-end ML pipeline to classify complex text patterns across a 200,000+ record dataset, leveraging 60,000-dimensional TF-IDF sparse features for rich semantic representation.
- Surpassed baselines to achieve a 0.826 Macro F1 score on a Kaggle-style benchmark by designing a Stacking Ensemble of 4 diverse linear models with a Logistic Regression meta-learner.
- Mitigated extreme class imbalances (21:1 ratio) and reduced overfitting through rigorous k-fold cross-validation and targeted class-weight optimization, deliberately avoiding noisy synthetic oversampling

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, SQLite), C/C++, JavaScript, HTML/CSS
Backend & Systems: Flask, REST APIs, SQLAlchemy, Redis, Celery, JWT (Auth & RBAC), Multi-tenant Architecture
AI/ML Libraries & Tools: Scikit-learn, Hugging Face (Transformers, Spaces), NumPy, pandas, KeyBERT, Gradio
AI/ML Techniques: Machine Learning Algorithms (Ensembles, SVM, Regression), Feature Engineering (TF-IDF, Dimensionality Reduction), Model Evaluation (k-fold CV, Bayesian Optimization), NLP, Semantic Matching, Statistical Analysis
Tools & Platforms: Git, Docker, Linux, AWS, Vercel, Supabase, VS Code, Jupyter Notebook

EXTRACURRICULAR & CERTIFICATIONS

DSA & Problem Solving: 850+ Day Coding Streak on GeeksforGeeks; solved 200+ LeetCode problems focused on algorithms, dynamic programming, and graphs.
Certifications (AI/ML): Machine Learning Specialization (DeepLearning.AI), NLP with Transformers using Hugging Face (IIT Madras), Machine Learning (NPTEL, IIT Kharagpur)